

RUDRA PATEL

+1 (352) 757-9258 patel.rudra@ufl.edu linkedin Portfolio

EDUCATION

University of Florida, Gainesville, FL	Aug 2025 – May 2027
M.S. Artificial Intelligence Systems	GPA 3.83/4.0
Parul University, Vadodara, India	Aug 2020 – May 2024
B.Tech. Computer Science (Internet of Things)	
Coursework: Operating Systems, Data Structures & Algorithms, OOP, Linear Algebra, Image Segmentation, Data Science, Machine Learning for IoT, NLP.	

TECHNICAL SKILLS

Languages	ML / Deep Learning	LLMs / NLP
Python, SQL, C++, Go, R, Bash	TensorFlow, PyTorch, Keras, PyTorch Lightning	Hugging Face, LangChain, RAG, FastAPI
Data & Visualization	MLOps / Tools	Core Concepts
Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, OpenCV	Docker, Git, GitHub, MLflow, W&B, Jupyter, VS Code	Data Pipelines, ETL, EDA, Feature Engineering, Model Training/Eval, Computer Vision, Statistics

EXPERIENCE

Graduate Research Assistant	June 2026 – Present
University of Florida	Gainesville, FL
- Annotated and quality-controlled 300+ whole-slide images for kidney and cancer pathology model training (HuBMAP, KPMP). - Automated gigapixel WSI preprocessing pipelines on HPC, cutting data-prep time by ~65% across 500+ slides. - Trained ResNet50/ViT models on the 2.3M-cell Immunocto dataset to subtype tumor-infiltrating immune cells from H&E (target 0.95 F1). - Benchmarked models against HoVer-Net and YOLOv10 baselines, analyzing F1, recall, and cross-cohort generalization. - Built reproducible, containerized ML workflows (Docker, Git) and contributed to lab tools FUSION and HistoLens.	

AI Automation Engineer	May 2024 – July 2025
Jignect	Ahmedabad, India
- Built Python automation and data validation pipelines, cutting regression testing time by ~35% across multiple releases. - Integrated ML-based validation checks into testing workflows, enabling earlier defect detection and improving pre-release data quality by ~25%. - Embedded automated checks in CI/CD pipelines, reducing deployment failures and production defects by ~20%. - Collaborated with cross-functional teams to standardize pipeline processes, accelerating release cycles by ~30%.	

Data Science Intern	December 2023 – March 2024
SlashMark	Hyderabad, India
- Designed data pipelines to ingest and preprocess 35,000+ retinal fundus images from the APTOS 2019 dataset for 5-stage classification. - Applied normalization and augmentation ETL steps at 224×224 resolution, reducing training loss by 28% over 25 epochs. - Trained a CNN with TensorFlow/Keras, achieving 96.3% training and 92.6% validation accuracy with Adam optimizer and early stopping.	

PROJECTS

End-to-End MLOps Pipeline	GitHub Python, MLflow, Docker	Jan 2026 – Feb 2026
- Engineered an end-to-end data pipeline automating ingestion, preprocessing, and model training, reducing manual workflow time by 45%. - Built modular ETL workflows with Docker containerization and CI/CD, enabling reproducible deployment and reducing environment errors by 40%. - Implemented experiment tracking with MLflow across 200+ runs, improving model reproducibility and pipeline reliability by 35%.		
Customer Segmentation Analytics	GitHub Python, Scikit-learn, Pandas	Sep 2025 – Nov 2025
- Performed exploratory data analysis on 50,000+ customer records to surface data-driven behavioral patterns. - Engineered feature pipelines and applied K-Means clustering, segmenting customers into 5 actionable groups at a 0.62 silhouette score. - Structured transactional data into analytical data models, improving reporting query efficiency by 30%.		
Time-Series Forecasting	GitHub Python, Pandas, Scikit-learn	May 2025 – July 2025
- Designed data pipelines to clean and transform 10,000+ time-indexed records for forecasting workloads. - Built and compared ARIMA and ML models, improving forecast accuracy (MAPE) by 22% over baseline. - Automated forecast reporting and feature extraction, cutting manual analysis effort by 40%.		
Customer Churn Prediction	GitHub Python, Scikit-learn	Feb 2025 – Apr 2025
- Developed ETL workflows to process 15,000+ customer records, handling missing values and class imbalance. - Engineered features and trained classification models, achieving 89% accuracy and 0.91 ROC-AUC to enable data-driven retention.		